

GenAI & Agentic AI Architecture

Architecture and Data-Focused · Omer Farooq

FOUR AI COMPONENTS

Data AI quality is bounded by the quality, relevance and accessibility of its data.	ML Learns patterns to predict, detect, rank and recognize.	GenAI Foundation models generate or transform text, code, images and audio.	Agentic AI Perceive, plan, act, observe: a loop with tools and memory.
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RAG = retrieve relevant context, augment the prompt, generate a grounded response.

MAP A USE CASE TO ITS COMPONENT

USE CASE	COMPONENT	WHY
Anomaly detection	ML	Learns normal patterns and flags deviations.
Recommendation	ML	Ranks items using learned user and item embeddings.
Computer vision	ML	Localizes pixel patterns without generation or planning.
Optimization	ML	Becomes agentic when embedded in an execution loop.
Autonomous planning	Agentic	LLM reasoning, tools and memory in a plan, act, observe loop.
Multimodal understanding	GenAI	Produces language responses from images, charts and video.

CROSS-CLOUD SERVICE MAP

SERVICE	OPEN SOURCE	GCP	AWS	AZURE
AI/ML platform	Kubeflow, MLflow	Vertex AI	SageMaker	Azure ML
Generative AI	LangChain, LlamaIndex	Gemini Enterprise	Bedrock	Azure OpenAI, Foundry
Agent orchestration	LangGraph, CrewAI	Agent Builder	Bedrock Agents	AI Agent SDK
Vector search	Milvus, Qdrant	Vertex Vector Search	OpenSearch Serverless	Azure AI Search
Document AI	Tesseract, LayoutLM	Document AI	Textract	AI Document Intelligence
LLM guardrails	Llama Guard, NeMo	AI Safety Filters	Bedrock Guardrails	AI Content Safety